HARVARD Kennedy School
JOHN F. KENNEDY SCHOOL OF GOVERNMENT

HKS Case 2174.0
March 23, 2020

The Chiapas Puzzle

For everyone, everything. For us, nothing.

– Slogan of the Zapatista Army of National Liberation

Introduction

In the early morning of December 15, 2013, Luis Videgaray was staring at the latest estimations of economic growth made by the National Institute of Statistics and Geography (INEGI). A year ago, Videgaray had been appointed Secretary of Finance and Public Credit by Mexico's incoming President Enrique Peña Nieto (2013-2018), and entrusted with the difficult mission to keep macroeconomic stability, accelerate growth, and at the same time reduce the large income gaps between northern and southern states. He had a rare combination of academic and political acumen, having earned his Ph.D. degree in Economics from the Massachusetts Institute of Technology (MIT) in 1998, and performed as Secretary of Finance for the State of Mexico (2005-2009, under then Governor Peña Nieto), Congress Representative (2009-2011), and Campaign Manager for Peña Nieto in the presidential elections of 2012. Now all of these skills were tested by Mexico's lackluster growth and mounting disparities between its affluent north and backward south, a situation that was far from new but – having worsened through the 1990s – was now threatening the political stability of the nation.

Over the previous decade, and in spite of having undertaken and completed a series of market reforms that had stabilized its macroeconomy, Mexico's growth rate had fallen well behind other developing nations and, in particular, Latin American counterparts. Between 2003-2013 the compounded annual growth rate (CAGR) of Mexico had been a squalid 1.3% per capita, third-from-bottom in Latin America above Guatemala (1.0%) and Haiti (0.1%). That was not the only puzzle Videgaray was facing. There was something even more disturbing on the INEGI report, which had to do with the growth statistics disaggregated by federal entities. Chiapas, Mexico's southernmost and poorest state (**Exhibit 1**), was also amongst the ones growing the least.[1] Over the previous twenty years

[1] Campeche was the only state with a CAGR lower than Chiapas. It was an oil-state that had witnessed over the previous decade the accelerated depletion of its main oil field. Cantarell, a giant off-shore oil field discovered in 1976, had registered a 79% volume loss between 2003 and 2013.

This case was written by Carolina Pan, Postdoctoral Research Fellow, Miguel Ángel Santos, Adjunct Lecturer in Public Policy, and Ricardo Hausmann, Rafik Hariri Professor of the Practice of International Political Economy, at the Harvard Kennedy School of Government (HKS). The authors want to thank Luis Videgaray, Dani Rodrik, Dan Levy, Luis Espinoza, Tim Cheston, and Patricio Goldstein for their useful comments and suggestions, and Christina Murphy and Anjani Datla at the Harvard Kennedy School Case Program/SLATE for their guidance and superb editing work. HKS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management. KS1307

Copyright © 2020 President and Fellows of Harvard College. No part of this publication may be reproduced, revised, translated, stored in a retrieval system, used in a spreadsheet, or transmitted in any form or by any means without the express written consent of the Case Program. For orders and copyright permission information, please visit our website at case.hks.harvard.edu or send a written request to Case Program, John F. Kennedy School of Government, Harvard University, 79 John F. Kennedy Street, Cambridge, MA 02138.

(1994-2013) Chiapas' income per capita had been falling at a CAGR of 0.7% (**Exhibit 2**). Contrary to the economic predicament according to which – under *certain conditions* – poorer places will tend to grow faster and eventually "converge" with the most affluent ones, Chiapas' income was diverging. All of the policy attention and resources poured in the state after the Zapatista uprising in January 1994, seemed to have resulted in a widening income gap. Even Guerrero and Oaxaca, southern states with significant indigenous populations – also targeted in the federal strategy to reduce the north-south income gap – had managed to grow at the same pace of Mexico over the previous fifteen years, keeping their income gap with the rest of the country relatively stable between 1998 and 2013 (**Exhibit 3**). Not Chiapas. Chiapas seemed immune to all efforts aimed at fostering convergence in income, even when they brought about convergence in multiple key indicators. Its income per capita gap with the rest of Mexico had widened steadily over the previous twenty years, going from 44% to 59% between 1994 and 2013. Unsurprisingly, by late 2012 Chiapas had the highest poverty (75%) and extreme poverty[2] (47%) rates in the country, as measured by both income and multi-dimensional measures.

As he browsed through the latest INEGI update, Videgaray noticed another bewildering fact: The inequality that persisted across Mexican states seemed to reproduce and amplify as in a fractal within Chiapas. While Mexico's richest federal entity (Distrito Federal) had an average income per capita six times higher than Chiapas, the wealthiest municipality in Chiapas (its capital, Tuxtla Gutiérrez) was about 8.5 times richer than the poorest (Aldama and Mitontic, **Exhibit 4**). In fact, Tuxtla Gutierrez' CAGR income per capita over the previous decade (4.4%) had tripled the Mexican average. Accordingly, poverty rates within Chiapas also displayed large differences, going from 13.1% in the capital to 80.2% in Mitontic and 83.2% in Aldama.[3]

That only made the whole puzzle even more complex – thought Videgaray – as the candidates to explain Chiapas' broadening gap would also have to account for the diverging trends observed within the state. What had gone wrong? What was different in Chiapas? Was the problem Chiapas, or was there something wrong with the *chiapanecos*?[4]

The Zapatista Rebellion

In the early morning of January 1, 1994 – on the same day the North American Free Trade Agreement became effective[5] – an army of approximately 3,000 indigenous people from the tribes Tzotzil, Tzeltal, Tojolabal, and Ch'ol, occupied the main squares of seven towns located in the highlands of Chiapas: San Cristobal de las Casas, Ocosingo, Las Margaritas, Altamirano, Chanal, Oxchuc and

[2] Poverty is defined as falling below a minimum threshold for income (welfare line) and lacking one or more social rights (i.e. health, education, social security, housing). Extreme poverty is a subset, with income below minimum welfare line and a shortage of at least three social rights.

[3] Poverty rates at the municipal level correspond to 2010.

[4] According to the Dictionary of the Spanish Royal Academy, chiapanecos is the demonym used in Spanish for the people from Chiapas.

[5] The North American Free Trade Agreement or NAFTA is a trade pact signed in 1992 and effective from January 1, 1994, aimed at creating a trade-free block by gradually eliminating tariff and non-tariff barriers on products and services traded among the USA, Mexico, and Canada.

Huiztán. The movement – presenting itself as the Zapatista National Liberation Army (EZLN, for its Spanish acronym) – declared war on the Mexican state and demanded land, jobs, housing, food, health care, education, independence, freedom, democracy, justice and peace.[1]

The masked rebels attacked civil centers, created barricades using furniture to defend their positions, and released more than 200 indigenous prisoners in San Cristobal de las Casas before the Mexican Army had any chance to react. A violent battle with government forces ensued, with the rebels holding on to their positions for a few days in some of these towns, and then retreating into the Lacandona jungle, where the fight was prolonged for a few more days. By the moment the Mexican government called for a ceasefire on January 12th, it is estimated that more than 300 people had died in the conflict.[2]

After the ceasefire, the Mexican government initiated a dialog with the Zapatista leaders that lasted for more than three years and led to the San Andres Agreement. According to the treaty, signed in the Chiapas town of San Andres Larráinzar on February 16, 1996, the Mexican government committed to eradicate "all political and social norms and forms generating or deepening the subordination, inequality and discrimination of indigenous people," while at the same time expanding their rights and autonomy over the occupied territories, "recognizing their cultural values, their right to live and enjoy their ancestral lands… the right to govern them using communal structures for self-determination, to develop their agriculture using their traditional production systems, and the right to define and execute their own development projects."[3]

The Mexican government established a commission to translate the San Andres Agreement into effective laws – the Commission for the Concord and Pacification of the Union – but the task proved easier said than done. On December 1996, after months of deliberations, President Ernesto Zedillo presented a proposal that departed significantly from the agreements, on the grounds that some tenets on the latter would have posed a risk for the union of the Mexican state.[4] In January 1997, representatives of the EZLN rejected the proposal, and refused to reopen negotiations.[5] From then onwards the dialog stagnated, and the social, cultural, and economic demands of the indigenous tribes that led to the Zapatista uprising were indefinitely postponed.

Chiapas: 20 Years after the Zapatista Rebellion

The audacious Zapatista uprising received massive international attention and put Mexico at the forefront of the struggle for justice and democracy in indigenous territories worldwide. Ironically, the same globalization forces that brought Mexico, Canada and the United States to sign NAFTA, made it possible for the Zapatista revolution to contest power, by appealing to social networks and universal rights at a global scale.[6] Even when the talks derived from the rebellion did not result in a concrete agreement – or probably because of it – Mexican governments from Zedillo onwards devoted significant attention and resources to the region, in an effort to address the large income gaps and prevent further social unrest.

Over the following twenty years, large investments in education, health, and infrastructure had been made, and a number of programs aimed at increasing the availability of credit for small and medium enterprises were launched. These policies, together with a massive conditional-cash-transfer program (*Progresa*) aimed at eliminating the "intergenerational transmission of extreme poverty," were the spearheads of the federal government's efforts to address poverty and stimulate economic activity in Chiapas.[7] By the end of 2012, it was Mexico's largest social program, both in terms of budget and beneficiaries, reaching 5.9 million households nationwide – 3.5 million of them in rural areas. This represented a total 25.7 million beneficiaries distributed over 2,400 municipalities. Chiapas was the state with the highest penetration, with almost universal coverage of poor families in rural areas, followed at a distance by the state of Mexico, Veracruz, and Puebla.

The challenges of deploying this package of social and economic policies were daunting. Chiapas is the eighth largest state in Mexico. It is traversed from northwest to southeast by two parallel mountain ranges, *Sierra Madre* and *Los Altos*, and two large rivers, *Usumacinta* and *Grijalva*, that produce highly heterogeneous regions in terms of climate and terrain. The state is fragmented into 122 municipalities, and by 2010 had 51% its population living in rural areas[6] – second only to Oaxaca's 53% among all Mexican entities. Only 19% of the people lived in the three municipalities with more than 100,000 inhabitants – Tuxtla Gutiérrez, Tapachula, and San Cristóbal de las Casas – versus a national average of 48% (see **Exhibit 5**). After Oaxaca (34%) and Yucatan (30%), Chiapas had the highest share of indigenous people in total population (27%),[7] concentrated around *Los Altos* and the *Lacandona* jungle. Most of the municipalities occupied by indigenous groups were ruled by *Usos y costumbres,* a system of self-government contained in the second article of the Mexican Political Constitution (1917), where a communal form of property – *ejidos* – prevailed. Communal property and the uncertainty derived from different *Usos y costumbres* – presumably a significant barrier to private investment in the area – was estimated to predominate in as much as 60% of total land in Chiapas.[8]

In spite of all these challenges, by December 2013 the impacts of all the policy attention and considerable resources poured into Chiapas by the federal government had left a very noticeable imprint in the state. Twenty years after the Zapatista uprising, all indicators appraising progress in the usual culprits of underdevelopment – the conditions considered necessary to buttress economic *convergence* in mainstream economics – had registered a significant improvement.

Education

By 2010 Chiapas had the lowest education attainment in Mexico. More than 60% of the economically active population (EAP) by 2010 had little to none schooling: 16% were illiterate, 21% did

[6] INEGI defines rural areas as localities – the social unit coming immediately after municipalities – with less than 2,500 inhabitants.

[7] Approximated by the number of people that declared being able to speak an indigenous language in the Population Census 2010. The five indigenous languages most important in Chiapas are Tzeltal (spoken by 37% of indigenous speakers), Tzotzil (34%), Chol (16%), Tojolabal (5%), and Zoque (5%). All of them, except the last one, belong to the Mayan family.

[8] Estimates by *Procuraduría Agraria, Programa de Certificación de Derechos Agrarios* (2012).

not complete primary school – more than twice than the (average) Mexican rates in both cases – and 23% did not finish secondary school.

There had been, however, significant investments in education after the Zapatista revolution. As a result, the number of primary and secondary schools had increased significantly since 1990. By 2010 Chiapas ranked first in the number of primary school establishments per capita, with over 177 per 100,000 people, and eighth in secondary schools per capita.[9] Consistently, the number of schoolteachers per capita was also among the highest in the country on these categories.

These investments had a significant impact, reducing the schooling gap by 47% in twenty years from an average of 3.2 years of schooling for the cohort born in 1965 to 1.7 for those born in 1987 (see **Exhibit 6**). Convergence in education was not mirrored by income: in 2010 the average Mexican worker, with 9.4 years of schooling and 22.2 years of experience,[10] was earning 83% more than the average worker in Chiapas, with 7.2 years of schooling and 22.7 years of experience. The difference in income between workers in Chiapas and those in the rest of Mexico persisted across all levels of education. That implies that in order to make the income of a worker elsewhere in Mexico, in Chiapas, you would need almost twice as much schooling (**Exhibit 7**).

There was also evidence suggesting that large income gaps were not driven by differences in the quality of education. First, according to the latest ENLACE (a country-wide standardized test in Spanish and Mathematics), by 2010 students in Chiapas ranked among the lowest in Spanish language, but above average in Mathematics.[11] Second, these large income gaps between workers with similar levels of education and experience almost vanished when a worker migrated out of Chiapas. By 2010, migrants from Chiapas working elsewhere in Mexico made an average income that was just 11% lower than other internal migrants in Mexico (**Exhibit 8**). When differences in the individual characteristics of migrants were considered, a worker born in Chiapas working elsewhere in Mexico was able to earn a salary that was just 2.7% lower than the average migrant with similar schooling, experience, gender, and indigenous origins (**Exhibit 9**). In spite of that, Chiapas registered the lowest internal migration rates amongst all Mexican states (2.28 per 1,000 inhabitants),[12] three times below the population-weighted average of the remaining Mexican states (6.18).

Health

Poverty and poor health tend to be very strongly correlated, and often times depicted as two sides of the same coin, also known as the health-poverty trap.[8] In the case of Chiapas, the large share of people living in rural areas as well as the linguistic barriers made health diagnosis and treatment

---

[9] There were 41 secondary schools per 100,000 people. The per capita number of universities was always low in Chiapas, which ranked 27th (of 32) in 2010.

[10] As it is common in the literature, experience here is calculated as age minus years of schooling minus six.

[11] A description of the ENLACE (National Evaluation of Academic Achievement in Education Centers) can be consulted here: http://www.enlace.sep.gob.mx/que_es_enlace/.

[12] Internal migrants are defined as workers born in Chiapas, that were working in a different Mexican state by the 2010 Population Census.

particularly challenging. Non-hospital settings, which were the most prevalent away from urban centers, lacked the proper equipment and relied on less precise diagnostic methods. In addition, government health care providers were mostly unfamiliar with indigenous languages, and there had been several reports of health care being denied to the indigenous peasants in the region.[9] And yet, while poverty in Chiapas remained high, health indicators improved considerably and started converging to the Mexican mean after 1995. In fact, Chiapas performed better in some indicators than the rest of the country.

By 2013 life expectancy in Chiapas was 72.5 years, lower than the national average of 74.5 but very close to that of other southern states. The overall mortality rate was 5.12%, 0.62 percentage points less than Mexico's average. The child mortality rate was 13.47%, mostly due to acute respiratory infections which had decreased significantly since 1990 but still had a much higher incidence than in other states. It is worth noting that Chiapas used to suffer from synergistic epidemics of tuberculosis in the early 1990s that led to high mortality rates, but those statistics had fallen from 16.5% (1990) to 2.88% (2013), outperforming many other states. Finally, other common mortality causes, like pulmonary arterial hypertension or cancer, were relatively low and in line with the national averages.[10]

Public spending in health services as a share of GDP was around 3% in Mexico, which is low for international standards.[13] In Chiapas, however, health expenditures represented 5.8% of the state's GDP, and were paid essentially by the federal government.[14] There were about 2,000 health establishments, 90% of which were non-hospital centers (versus 78% for the country). High-technology medical equipment is notably scarce for the population count, with less than 35 devices across the entire state, but this was a nationwide problem. The number of doctors per capita was also not far from the national average.

Access to Credit

Banking penetration and access to credit have remarkably improved in Chiapas, and did not look different than in the rest of Mexico. According to the Economic Census of 2008, 30% of the business units registered in Chiapas had access to some form of financing, slightly above the national average (28%). When restricted to external financing – excluding shareholder funds and retained earnings – that number fell to 14%, slightly below the national average (15%).

Between 2009 and 2013, in the aftermath of the global financial crisis, interest rates had declined at the national level. In Chiapas, real interest rates had been 0.3 percentage points above the national average for micro and small firms, below average for medium and large companies by 0.2 and 0.9 percentage points respectively (**Exhibit 10**). By the end of 2013, for any company size, interest rates in Chiapas were among the lowest in Mexico (see **Exhibit 11**). In any case, real interest rates were not very different across Mexican states, a feature that is consistent with the free mobility of capital within the

[13] The OECD average was 6.5% in 2013.

[14] According to Direccion General de Informacion de Salud (DGIS), the share of health expenses in Chiapas paid by the federal government went from 89% in 2003 to 99% in 2013.

union. In that context, projects compete nationwide to attract the available funds, regardless of where businesses are located and savings are generated.

Some features in the credit markets were particularly distinctive in Chiapas. By 2010 it was the state with the third largest share of consumption and mortgage in overall credit. Conversely, commercial credit as a percentage of non-oil GDP was just 3.8%, fourth from the bottom after Hidalgo, Oaxaca and Guerrero (**Exhibit 12**). As a result, the composition of credit in Chiapas departed significantly from the national average. In particular, the share of credits to consumption in total credit in Chiapas was 3.6 times higher than the national average (36.4% vs. 10.1%); whilst the share of mortgage credit was 7.0 times higher (6.4% vs. 0.9%). Those higher shares came at the expense of commercial credit, and in particular productive credits, which accounted for 34.9% of total credit in Chiapas, in contrast to 65.1% at the national level.[15]

Electricity

Access to electricity in Chiapas had jumped from 78% to 94% of total households between 1995 and 2010, reducing the gap with the average of the remaining Mexican states from 15 to 2 percentage points in fifteen years.[11] By 2013, Chiapas was the third largest net energy producer state in Mexico, generating over four times the amount of energy consumed (**Exhibit 13**). With energy consumption growing at a rhythm of 2.8% per year, saturation seemed very unlikely even in the medium run. In Mexico, prices were set by a state-owned company with monopoly power. While there was price discrimination according to different categories of consumers (domestic, agricultural, industrial, commercial and utilities), the rates were the same throughout all states.

While this pricing regime could not possibly explain differences in performance across sectors in Mexico, it prevented Chiapas from benefiting from its natural competitive advantage, and in practice operated as a cross-subsidy going from energy-abundant to energy-scarce regions.[12]

Infrastructure: Roads, Airports, and Ports

Infrastructure – or the lack thereof – is always at the center of attention when it comes to identifying the culprits of underdevelopment. In the case of Chiapas, that emphasis was paramount given its very particular characteristics: rugged geography, location at the southernmost point of Mexico, political and administrative fragmentation, and large share of population living in rural areas.

Indeed, one of the first letters commending Videgaray for being appointed Secretary of Finance and Public Credit on December 2012 had come from Ovidio Cortazar Ramos, the Secretary of Finance of the State of Chiapas. In the text of the letter, he stressed the importance of working together to produce a breakthrough in Chiapas and described what he considered to be the priorities of a new development agenda for the state:

[15] Commercial credit includes productive credit (used for production purposes in agriculture, livestock, industries and services), credit for financial sector (both private and public), and to public sector.

> Our characteristics pose a significant challenge to connectivity, limiting – in time and distance – the transportation of people and products… In particular, when it comes to roads, the emphasis is often placed on the infrastructure connecting Chiapas with the rest of Mexico, and not enough attention is placed on the state of roads connecting communities and markets within the state. That is where the largest needs of infrastructure are; even more when you consider the hardships of building and maintaining infrastructure in the presence of factors such as orography, weather, constant rainfall and meteorological phenomena, such as hurricanes and earthquakes.[13]

The improvements in connectivity between Chiapas and the rest of Mexico implicit in Cortazar's letter were a relatively recent development. In the early 2000s, it was widely believed that issues with connectivity were at the core of the large income differences registered between northern and southern Mexican states. In particular, Davila, Kessel and Levy had argued that the fact that most roads in the South were made to connect directly with Mexico City was causing congestion and increasing transport times in and around the capital, as goods were forced to go through the city when shipped either internally or internationally. The authors suggested a number of infrastructure developments, mostly highways and coastal roads, to improve north-south connectivity.[14] A decade later, most of the proposed projects had been completed, and their estimates of savings in distance and times had been surpassed.[15] Yet another example of convergence in development inputs that was not translated to income.

Among the most significant investments completed was a highway system that went all the way up to the border with the United States in Nogales, which reduced transportation time from Tapachula – Chiapas southernmost urban center – by 33% or 13 hours. Chiapas connections to the outside world had also been enhanced by massive infrastructure investments in airports and ports. By 2012, Chiapas had two functioning commercial airports, located at Tuxtla Gutierrez and Tapachula, and a third one was about to open in Palenque.[16] In addition to that, by virtue of an investment of more than a billion dollars, the Chiapas Port had been upgraded to a full cargo port and cruise liner dock. While the rationale behind this massive project was to foster exports, restrictions on cargo load and single-port-entry limited the usage of the port that functioned at a minimal capacity.

With regard to Cortazar's claims that road infrastructure within Chiapas was a major binding constraint to growth and development, some evidence seemed to suggest the opposite. When assessing road network quality considering population and area, Chiapas did not underperform relative to the rest of the country either from a geographic or demographic standpoint. By 2013, Chiapas had 13.2 kilometers of paved roads per 10,000 inhabitants, and 9.2 kilometers per 100 square kilometers, far from the leading states but well above the Mexican average (**Exhibit 14**).

[16] The airport at Palenque opened in 2014, and its administration was given in concession to *Grupo Aeroportuario Chiapas*. Five years later it was handling 17,500 passengers per year, with four flights per week connecting Palenque with Mexico City and Monterrey. Although it was upgraded to receive international traffic in 2016, at the time of writing no agreement had been made to realize that possibility.

Business Climate

According to the World Bank's Doing Business Indicators,[16] the business climate in Chiapas by the end of 2013 was relatively good. That year Mexico ranked second only to Colombia in Latin America for ease of doing business, whilst Chiapas ranked 5th amongst its 32 states. In addition to that, in the only two corruption indicators available at the subnational level,[17] Chiapas ranked well above the Mexican average. When it came to security – a major issue in Mexico mostly related to drug cartels – citizens of Chiapas did not perceive the state as insecure.[18] Moreover, in terms of crime and murder statistics, Chiapas ranked among the safest Mexican states.

The Economic Structure of Chiapas

If anything had not changed much in those twenty years it was the economic structure of Chiapas and its sharp contrasts to that of Mexico's (**Exhibit 15, upper panel**). *Agriculture, Forestry, Livestock and Fishing* were the main sources of employment in Chiapas, albeit their importance had somewhat declined from 58% (1990) to 44% (2010). *Wholesale and Retail* – the second most important employers throughout – had picked some of that lost share, going from 8% to 12% between 1990 and 2010. By 2010 the third most important employer was *Construction*, with a share of 8% (5% in 1990). Overall, the changes recorded in employment were a mirror of what had happened with the state at large: a steady shift from tradable goods to non-tradable services, financed by massive transfers of federal resources.

In the rest of Mexico, *Agriculture, Forestry, Livestock, and Fishing* was also the principal employer, but with a share of total employment that was much smaller than that of Chiapas, and had shrunk at a much faster pace over the previous twenty years – going from 22% in 1990 to 13% in 2010. The second and third largest employers alternated between *Manufacturing, Wholesale, and Retail*, with the former going from 19% in 1990 to 15% in 2010 (three times higher than Chiapas), and the latter moving inversely from 13% in 1990 to 19% by 2010 (55% higher than in Chiapas). Overall, the economic structure of Mexico was more modern and diversified than that of Chiapas (**Exhibit 15, lower panel**).

When it comes to public employment, the aggregated employment shares of *Government activities*, *Education* and *Health and Other services* (as these two have a significant component of public employment) suggest that Chiapas had been gradually catching up. Their combined employment share went from 9% in 1990 to 13% in 2010, a number very close to the one exhibited by these categories at the national level throughout (12% - 13%).

But the large income gaps were not entirely driven by different sectoral compositions. When it came to productivity – as proxied by wages – there were large differences within sectors between Chiapas and

[17] "Índice de calidad de la transparencia", published by Centro de Investigación y Docencia Económicas (CIDE, "Center for Research and Economic Teaching"), and "Índice de corrupción y buen gobierno" (corruption and good governance index) made by Transparencia Mexicana.

[18] In the "perceptions of insecurity" index (published by Instituto Ciudadano de Estudios Sobre la Inseguridad), Chiapas ranks as the tenth state with the smallest perception of insecurity in the country.

the rest of Mexico (**Exhibit 16**), which had widened in time. In *Agriculture, Forestry, Livestock and Fishing*, for instance, the gap between the wages of workers in Chiapas relative to the same sector in the rest of Mexico had gone from 17% in 1990 to 46% in 2010. A similar trend could be observed in *Wholesale and Retail* – where the wage gap expanded from 16% to 33% - and *Manufacturing* (from 38% to 49%) over the same time period.

One Hippo in the Desert

Government efforts to supply the complementary inputs required to spark growth and development had only produced one, very noticeable appearance within the traditional landscape of the Chiapas economy: Yazaki. The Japanese multinational assembled electric harnesses, connectors, and other components of car boards, which were entirely exported to the United States and Japan. The company had decided to move their manufacturing plant from Nuevo Leon to Chiapas in late 1994 – the same year of the Zapatista uprising.

The arrival of Yazaki was driven by an effort led by President Ernesto Zedillo to bring new, modern business into Chiapas. The company had been based in Nuevo Leon, but fast economic growth in the northern state had pushed wages upwards and squeezed its profit margins. They were a labor-intensive firm, that relied on cheap, primary-schooled labor and good road infrastructure connecting Chiapas with the port of Salina Cruz (Oaxaca) and the northern border with the United States. Those factors were not there by 1994. In order to facilitate the move, the federal government had given Yazaki a large plot of land in Tuxtla Gutierrez, improved the roads connecting the capital to major interstate highways, and subsidized the training of 600 workers on the assembly line for an initial period of three months. That had broken the chicken-and-egg dilemma: The know-how required was brought into the state by Yazaki from Nuevo Leon – a team of highly qualified engineers with significant experience in the firm – and the federal government supplied the public goods and investments required to jump start operations.

Twenty years down the road, Yazaki had expanded on its own into five different locations within Chiapas, and had two more in the pipeline (**Exhibit 16**). These plants employed more than 3,000 workers, and exported 79.1 million dollars.[17] That was in part eased by the fact that the assembly line of Yazaki could be broken into smaller pieces that demanded more manual work, which could be performed in different facilities and then shipped to large assembly lines in their headquarters at Tuxtla Gutierrez. Most of the team that had set-up the Yazaki operation in Chiapas had since gone back to Nuevo Leon, with their places being taken by engineers graduating from *Universidad Autonoma de Chiapas*.

In each facility the company operated its own transportation system for workers – a network of red buses which covered as much as a 30-kilometer radius around the plant. Among the Yazaki workers there were many of indigenous origins, that were picked up and returned daily from communities governed by *Usos y Costumbres* within the reach of its transport fleet.

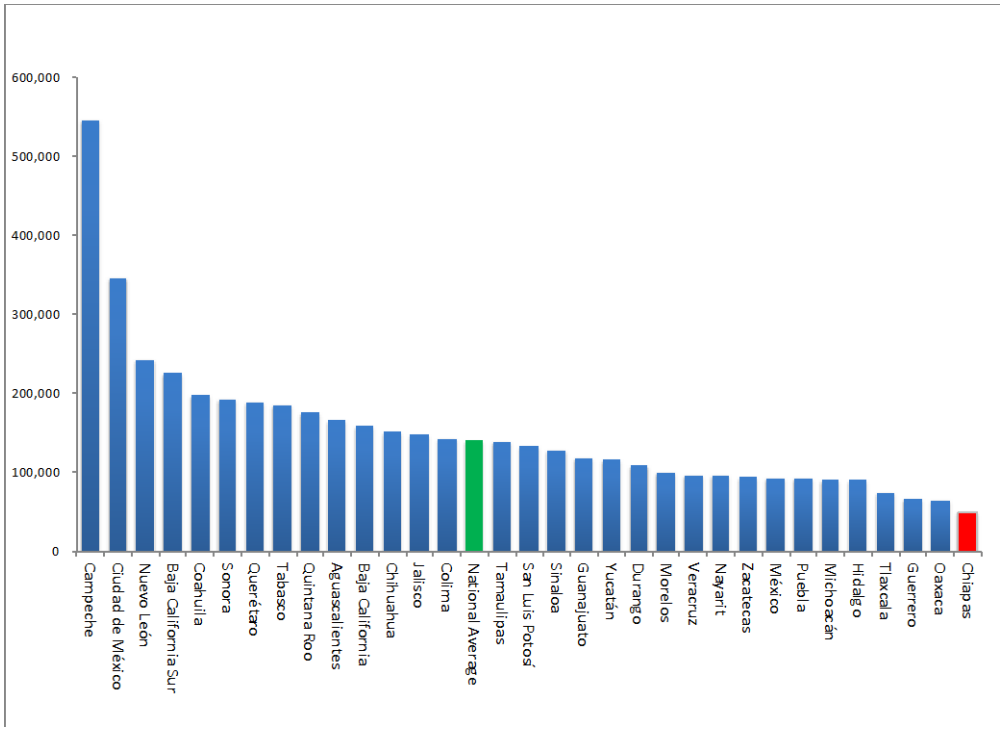
In spite of the geographically spread expansion, the company had not considered the highlands of Chiapas as a potential location for two main reasons. First, there communal property was the norm, a

feature that would prevent the company from renting or purchasing land. Second, social conflicts erupting every now and then, however minor, often led to roadblocks that threatened the timeliness of their shipments and ultimately their financial profitability.

Decision

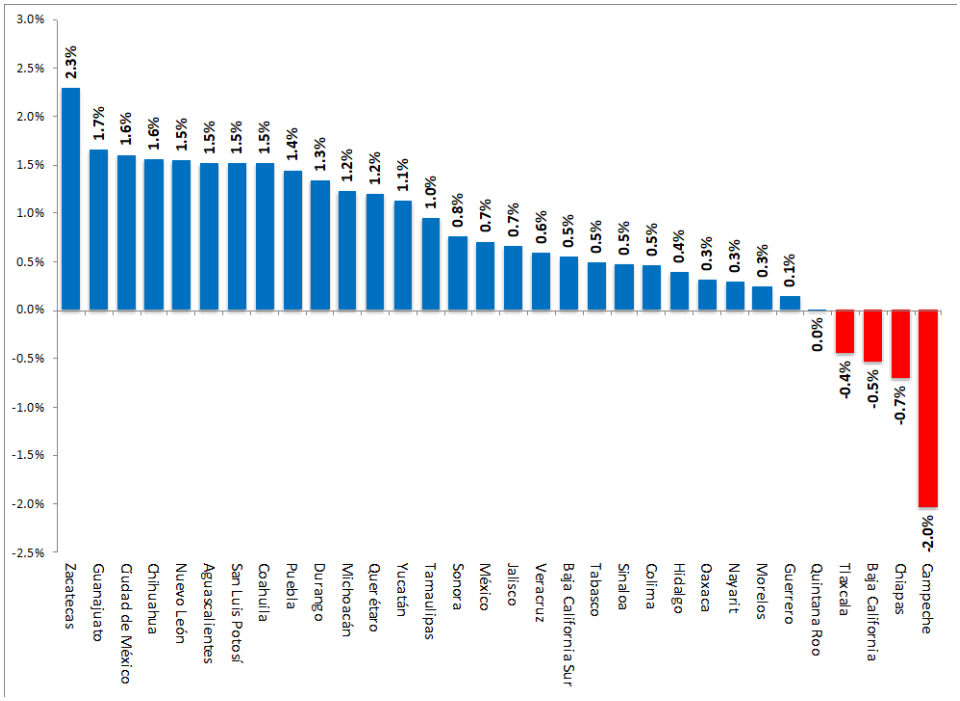
Videgaray finished browsing through the INEGI report, and leaned back on the chair gazing through the windows of his office at *Palacio Nacional.* His eyes stopped at the 330-foot pole at the center of the colossal *Plaza Constitución* – known to Mexicans as the *Zocalo* – and the no less imposing, massive waving Mexican flag first set there by President Ernesto Zedillo in 1999. What could be done to spark economic growth and development in Chiapas, he pondered, that had not been done already? Why hadn't all the investments and improvements made in those twenty years in education, health, infrastructure, access to energy and governance, bring the incomes of Chiapas' workers closer to the national average? What else could be done to reduce the large productivity gaps between workers in Chiapas and those in the rest of Mexico? Why in twenty years had no other modern manufacturing firm followed Yazaki, in spite of the evident success of their operation there?

Exhibit 1: Net Value Added Per Capita by Federal Entity 2013
(constant pesos 2013)



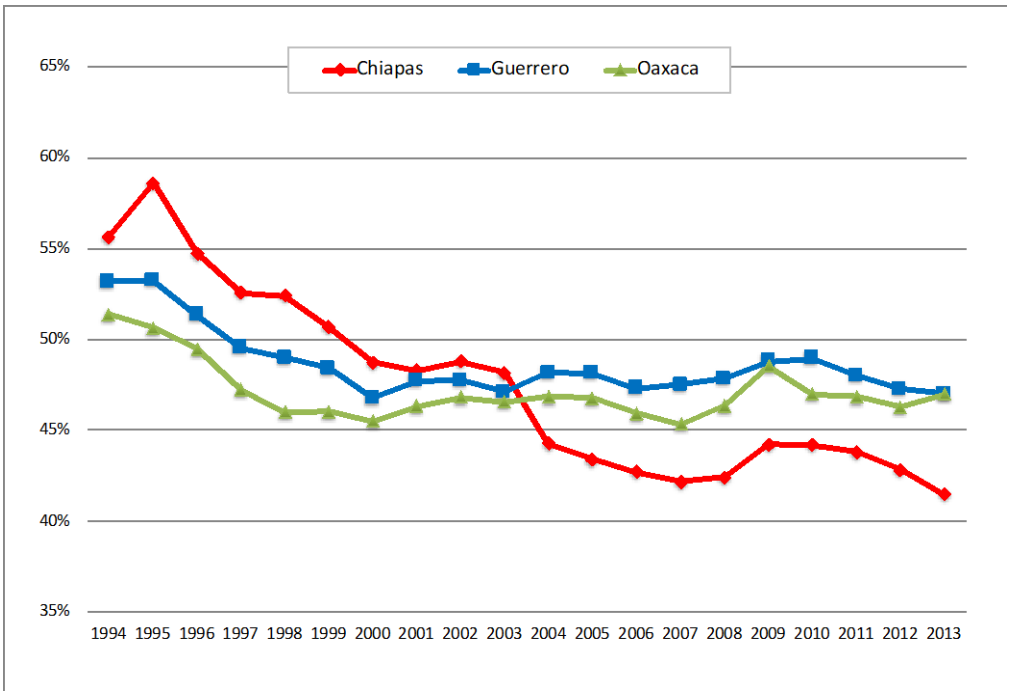
Source: INEGI.

Exhibit 2: Compounded Annual Growth Rate Per Capita (1994-2013)



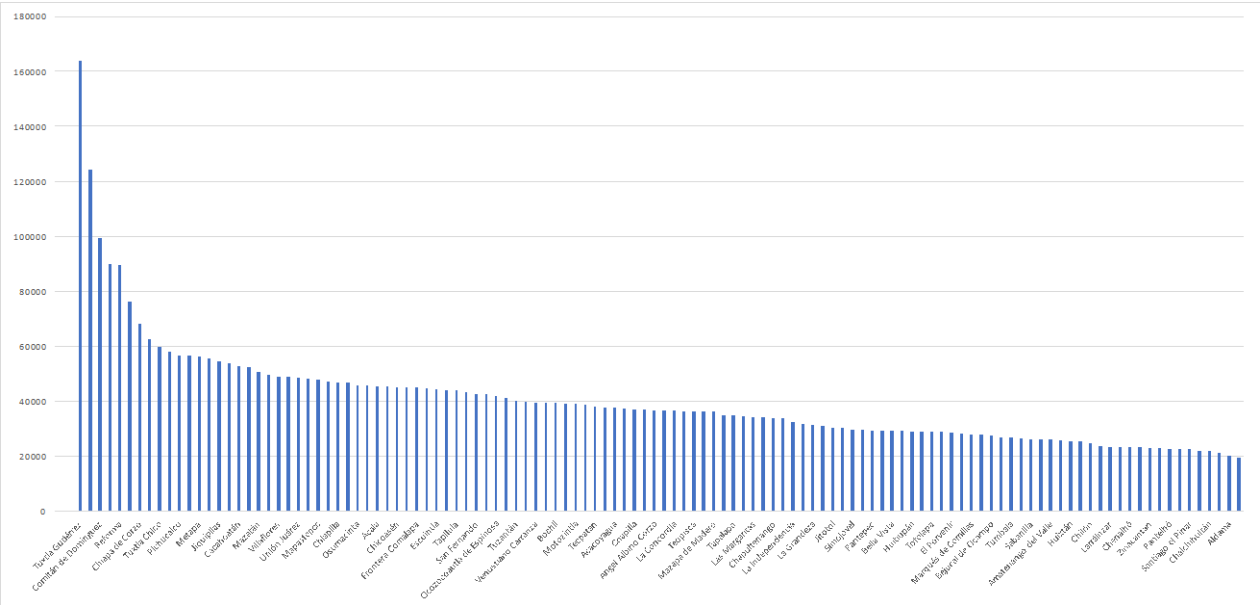
Source: INEGI.

Exhibit 3: Net Value Added Per Capita
(as percentage of average national income)



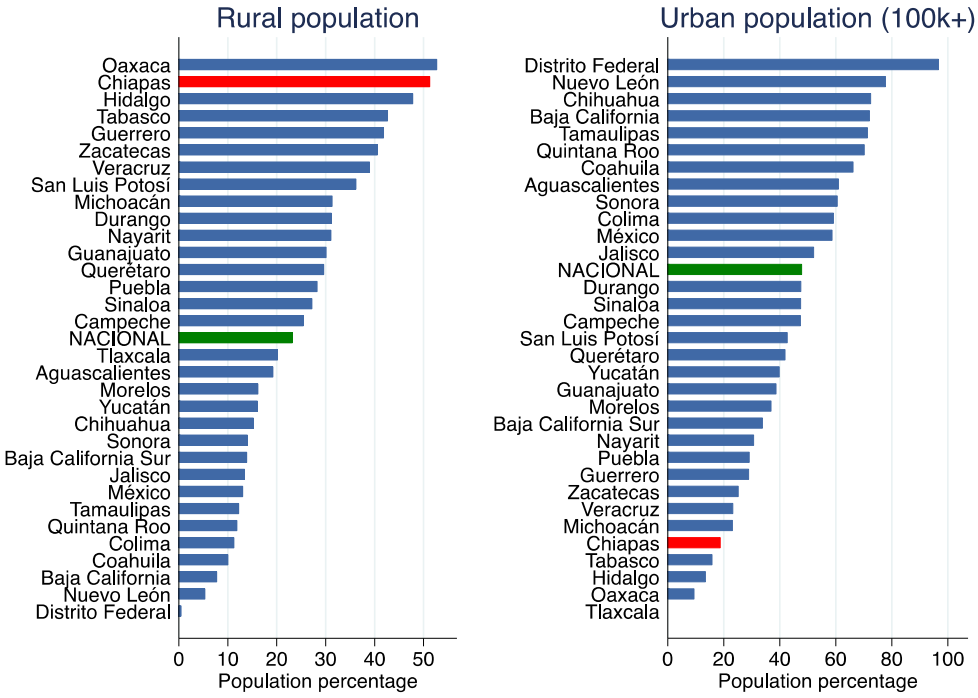
Source: INEGI, CONAPO, authors' own calculations.

Exhibit 4: Non-Oil GDP Per Capita by Chiapas' Municipality
(constant pesos 2010)



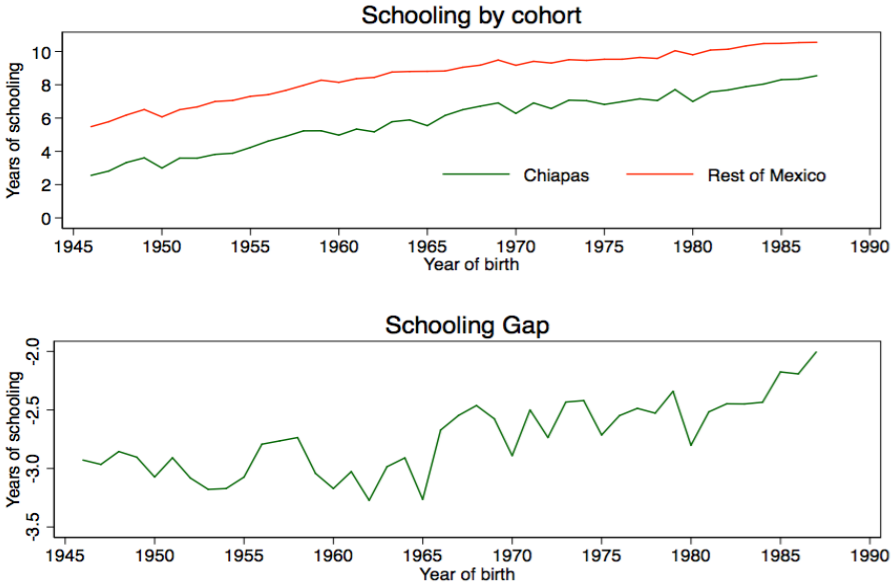
Source: INEGI.

Exhibit 5: Population Distribution in Rural and Urban Localities of More Than 100,000 Inhabitants



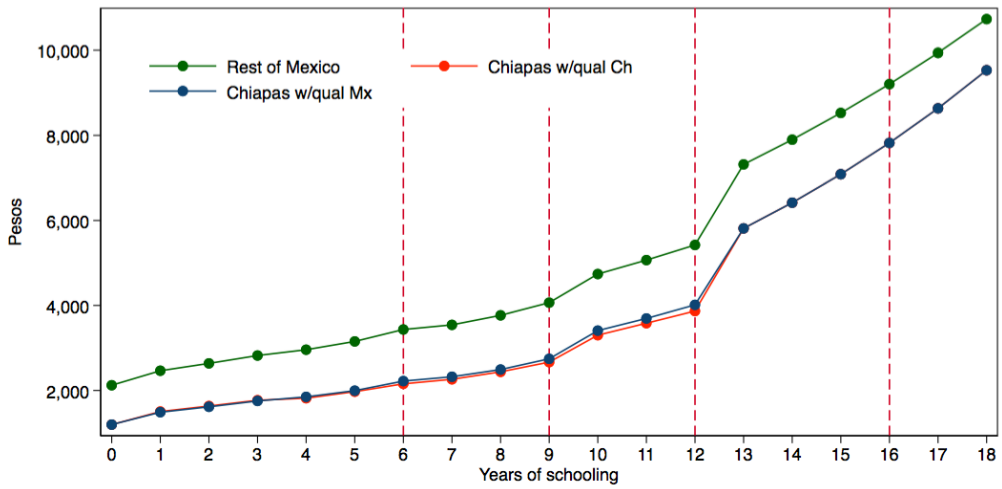
Source: INEGI, Censo de Población y Vivienda (2010).

Exhibit 6: Education and Gap with Mexico by Year of Birth (Chiapas, 2010)



Source: 2010 Population Census (INEGI).

Exhibit 7: Labor Income by Years of Schooling



Source: 2010 Population Census (INEGI) and ENLACE scores (Secretaría de Educación Pública).

Description:

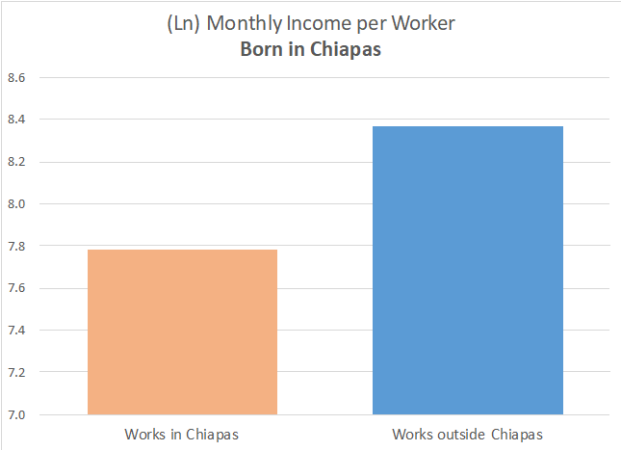
Author's calculations of expected income of the average worker in Chiapas and Rest of Mexico based on a Tobit regression (for each of the two) of income on education and experience:

$$\hat{Y} = \widehat{\beta_0} + \widehat{\beta_1} educ + \widehat{\beta_2} exp + \widehat{\beta_3} exp^2 + \widehat{\beta_4} quality + \varepsilon$$

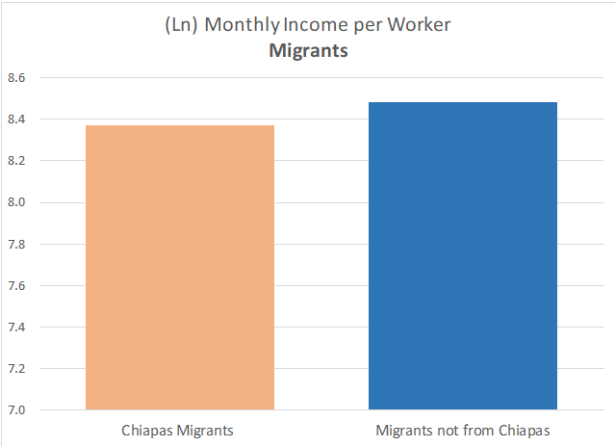
Years of education and experience are obtained from the 2010 Population Census. The quality of education is measured using ENLACE test scores. The blue line represents expected income in Chiapas if the average worker had the same education quality as in the rest of Mexico.

Exhibit 8: Migrants incomes (Chiapas vs. Rest of Mexico)

Panel a) Workers from Chiapas vs. Rest of México



Panel b) Migrants coming Chiapas vs. Rest of México



Source: Population census, 2010.

Exhibit 9: Tobit Regression of Income Per Worker and Migrants

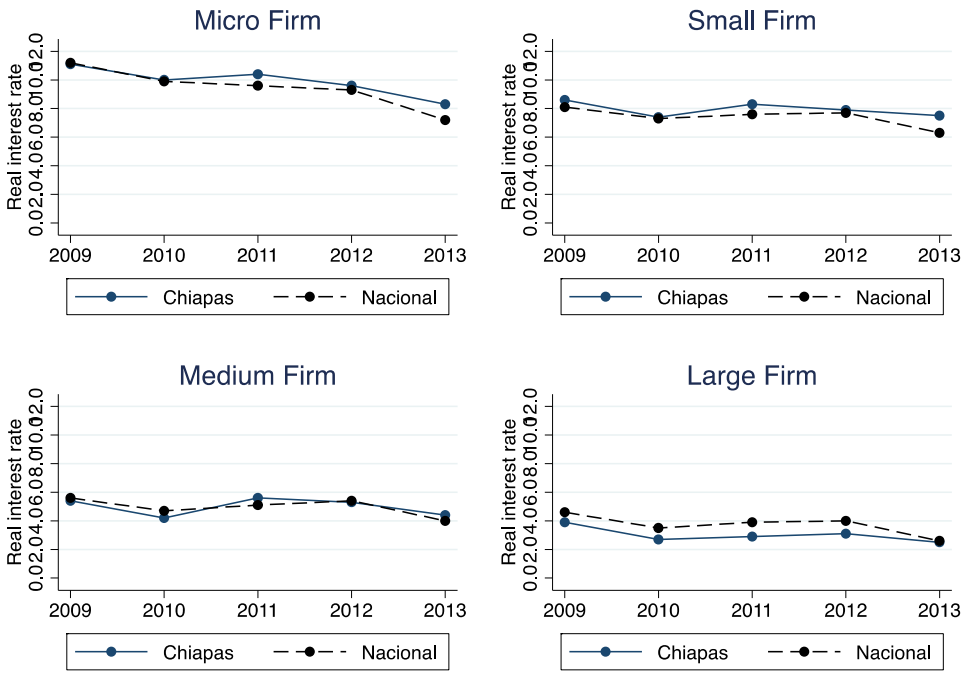
	(1)	(2)	(3)
Years of Schooling	0.095***	0.095***	0.095***
	335.17	335.06	325.79
Experience	0.032***	0.032***	0.032***
	310.78	311.13	310.98
Experience-squared	-0.000***	-0.000***	-0.000***
	-241.36	-241.16	-241.23
Female	-0.337***	-0.337***	-0.340***
	-258.12	-259.59	-266.92
Indigenous Language	-0.262***	-0.260***	-0.250***
	-33.45	-33.94	-46.64
Born in Chiapas	-0.269***	-0.346***	-0.406***
	-25.22	-27.01	-24.04
Migrant	0.139***	0.128***	0.128***
	66.68	61.21	61.23
Migrant*Chiapas		0.384***	0.371***
		23.74	27.57
Years of Schooling*Chiapas			0.005***
			5.12
Experience*Chiapas			0.000
			0.07
Female*Chiapas			0.102***
			9.06
Indigenous*Chiapas			-0.104
			-1.63
Constant	7.125***	7.126***	7.129***
	2042.65	2041.53	2004.13
Observations	3,005,931	3,005,931	3,005,931

t values are indicated beneath the coefficients.
*** p<0.01, ** p<0.05, * p<0.1

Description:

Author's calculations of expected income of the average worker in Mexico based on a Tobit regression of income from work, controlling for years of schooling, experience, gender, indigenous language, place of birth (Chiapas), migrant condition (works in state that is different from birthplace), and some interactions to account for incremental impacts of these variables in Chiapas.

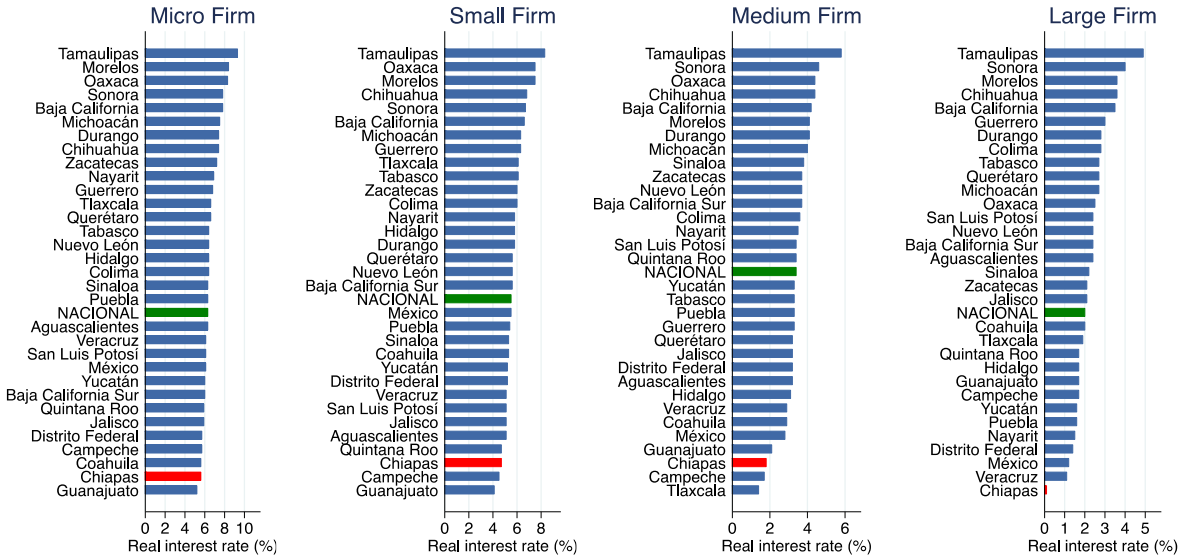
Exhibit 10: Evolution of Real Interest Rates by Company Size (2013)



Note: Real interest rates were obtained by subtracting from domestic interest rates the variation of the consumer price index by state.

Source: Comisión Nacional Bancaria y de Valores (interest rates) and INEGI (inflation by entity).

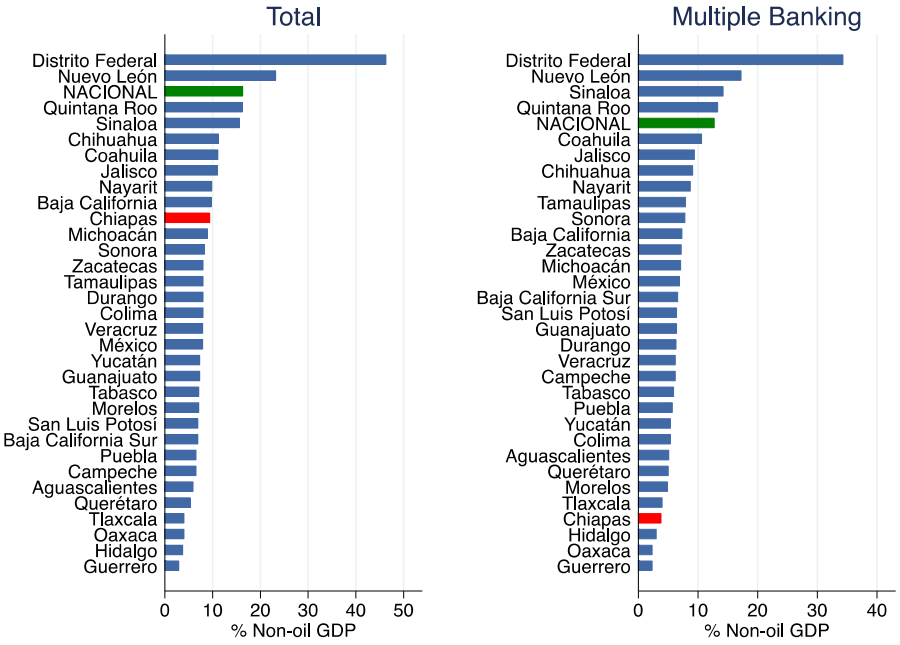
Exhibit 11: Real Interest Rates by State (2013)



Note: Real interest rates were obtained by subtracting from domestic interest rates the variation of the consumer price index by state.

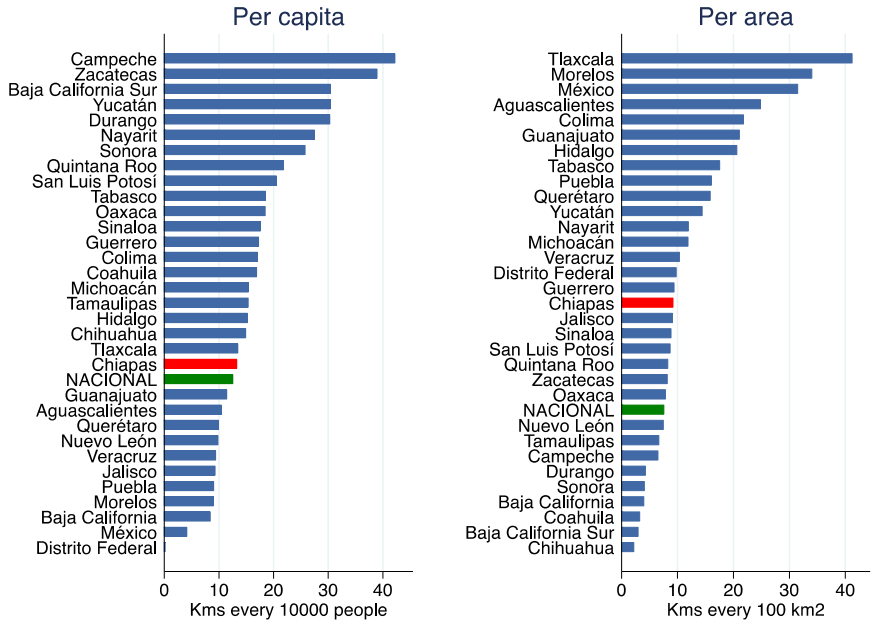
Source: Comisión Nacional Bancaria y de Valores (interest rates) and INEGI (inflation by entity).

Exhibit 12: Total Productive Credit and Credit from Multiple Banking (States, 2013)



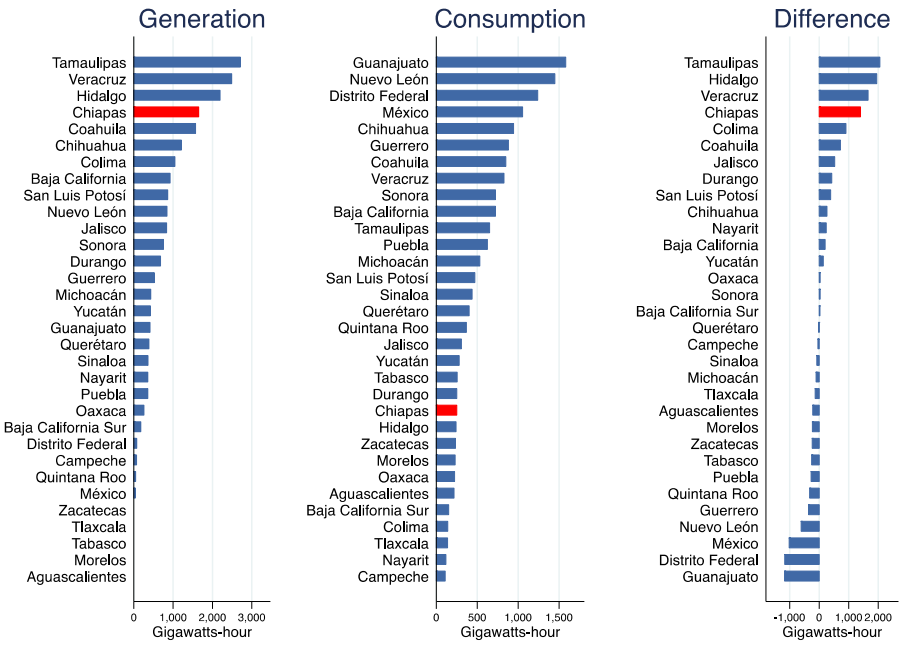
Source: Comisión Nacional Bancaria y de Valores (interest rates) and INEGI (inflation by entity).

Exhibit 13: Kilometers of Paved Roads per Number of People and Area (2013)



Source: Instituto Mexicano para la competitividad (IMCO).

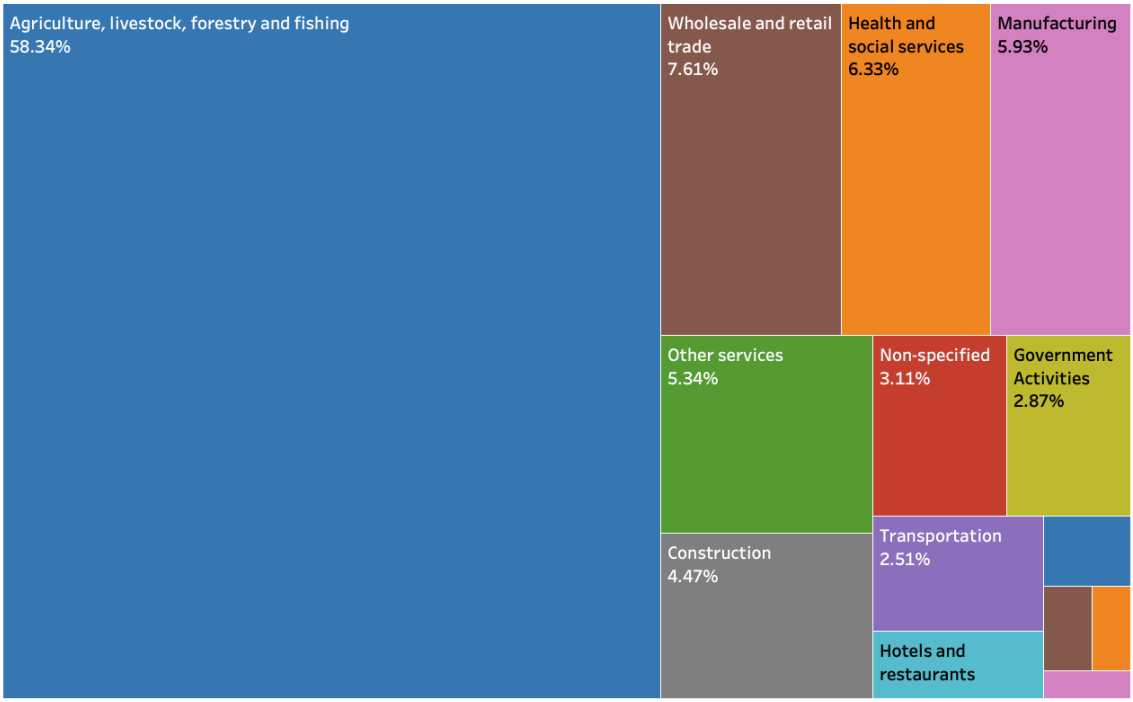
Exhibit 14: Electric Power Net Generation as Percentage of Consumption (2013)



Source: Comisión Nacional de Electricidad (CNE).

Exhibit 15 (a): Employment shares in Chiapas 1990-2010

Chiapas 1990



Chiapas 2010

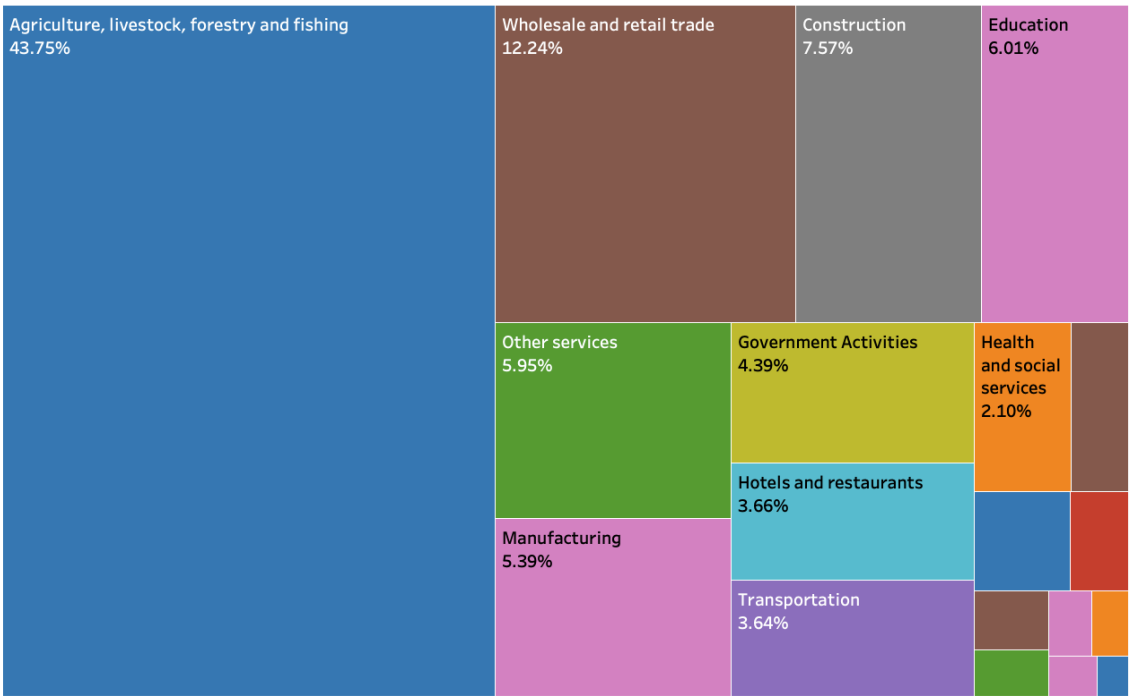
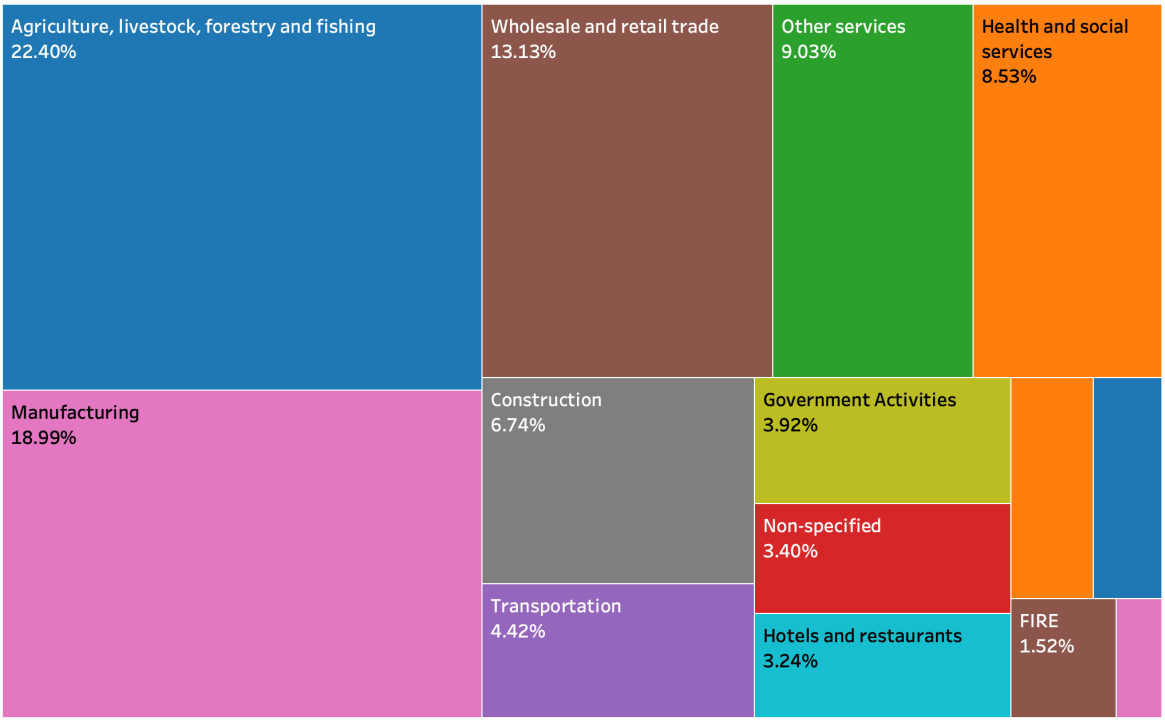


Exhibit 15 (b): Employment shares in Mexico 1990-2010

Mexico 1990



Mexico 2010

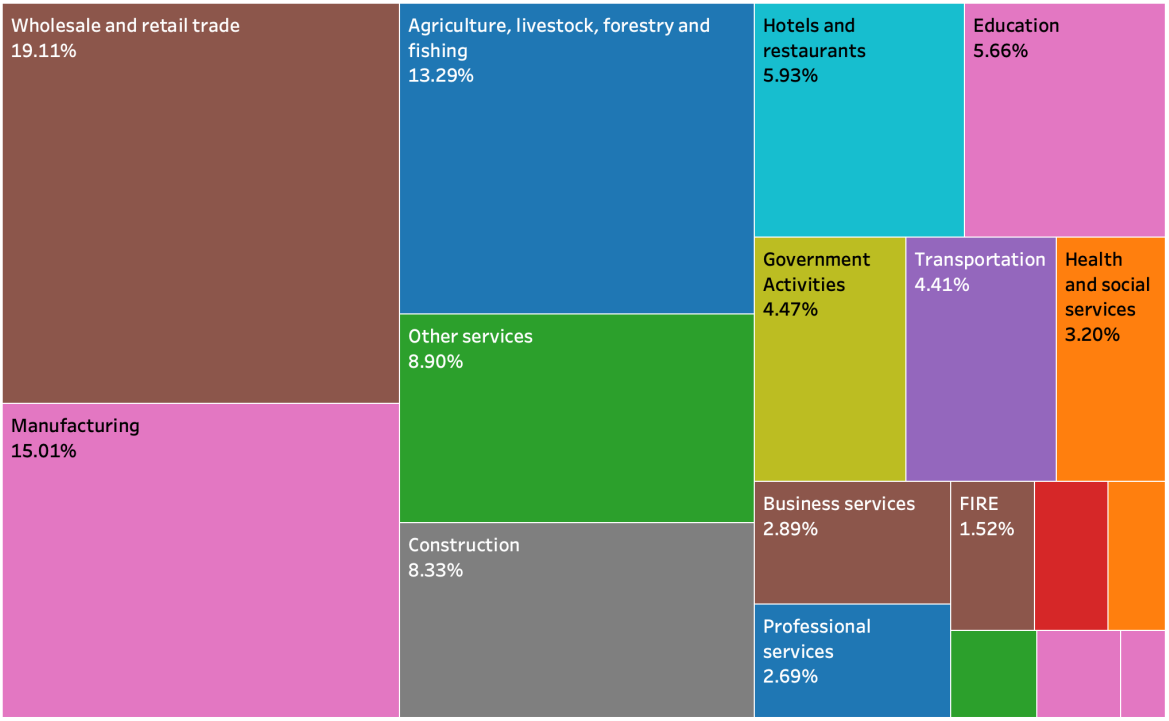
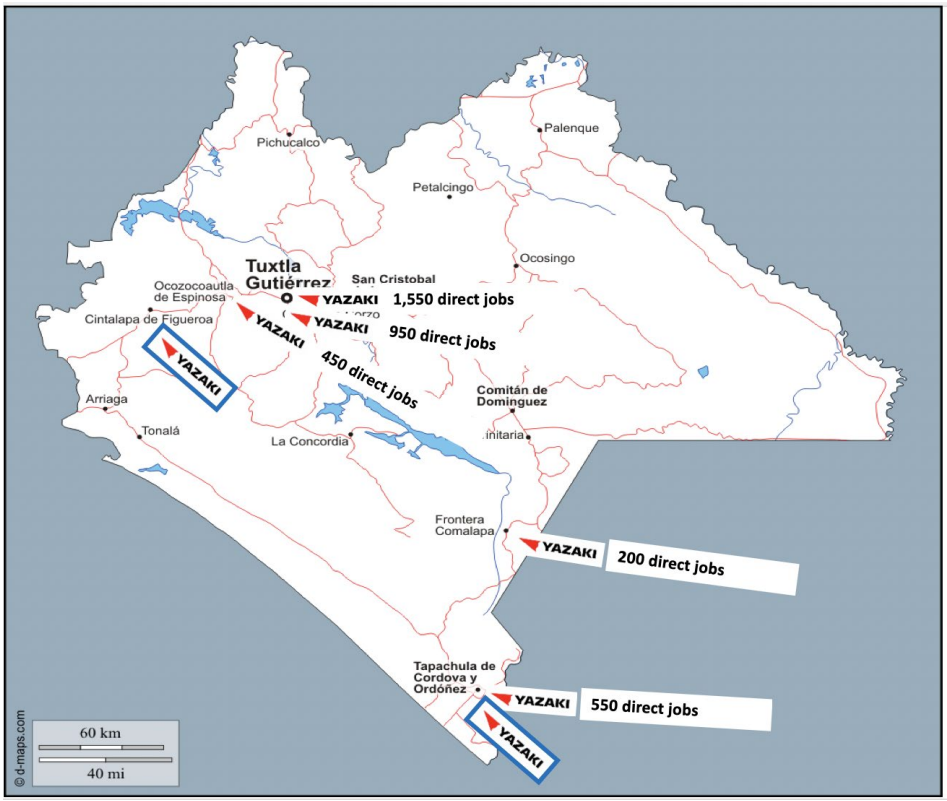


Exhibit 16: Relative Wages for Selected Sectors

Source: Population Census.

Exhibit 17: The Yazaki Footprint in Chiapas



Note: Squares in blue indicate two additional plants that were being planned by the end of 2013.

Endnotes

[1] Neil Harvey, *The Chiapas Rebellion: The Struggle for Land and Democracy*, Durham: Duke University Press, 2008.

[2] S.D. Collins, *Indigenous Rights and Internal Wars: The Chiapas Conflict at 15 Years*, The Social Science Journal, 2010.

[3] L. Hernandez Navarro, *Acuerdos sobre Derechos y Cultura Indígenas: Acuerdos de San Andrés*, Ediciones Era, 1998.

[4] Graciela Marquez, *Del autoritarismo agotado a la democracia frágil: Nueva Historia general de México*, Mexico: El Colegio de México, 2010.

[5] Luis Hernandez Navarro, Serpientes y Escaleras: Los Avatares de la Reforma Constitucional sobre Derechos y Cultura Indígenas, 1998.

[6] Richard Stahler-Sholk, "Globalization and Social Movement Resistance: The Zapatista Rebellion in Chiapas, Mexico," New Political Science, 2010.

[7] Santiago Levy, *Progress against Poverty: Sustaining Mexico's PROGRESA-Oportunidades Program*, Washington D.C.: Brookings Institute, 2006.

[8] Xavier Sala-i-Martin, "On the Health-Poverty Trap," in *Health and Economic Growth: Findings and Policy Implications*, Cambridge, MA: MIT Press, 2005.

[9] M.E. Herce, J.A. Chapman, A. Castro, G. García-Salyano, and K. Khoshnood, "A role for community health promoters in tuberculosis control in the state of Chiapas, Mexico," *Journal of Community Health,* April 2010, 35(2), pp. 182-189.

[10] Dirección General de Información en Salud (DGIS): http://sinaiscap.salud.gob.mx:8080/DGIS/.

[11] Sistema de Información Energética.

[12] E. Dávila, G. Kessel, and S. Levy, *El sur también existe: un ensayo sobre el desarrollo regional de México*, Economía mexicana. Nueva Época, 2002.

[13] Ovidio Cortázar, Secretary of Economic Development of Chiapas, "Chiapas: Insumos para el diseño de una nueva estrategia de desarrollo."

[14] E. Dávila, G. Kessel, and S. Levy, *El sur también existe: un ensayo sobre el desarrollo regional de México*, Economía mexicana. Nueva Época, 2002.

[15] R. Hausmann, L. Espinoza, and M.A. Santos, "Chiapas Growth Diagnostics: The Low-Productivity Trap," Harvard Center for International Development, Faculty Working Paper No. 304, October 2015.

[16] *Doing Business 2014: Understanding Regulations for Small and Medium-Size Enterprises,* Washington DC: World Bank Group, 2013.

[17] Mexican Atlas of Economic Complexity.